Yash Seetharaman

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Education

University of California, San Diego – Jacobs School of Engineering La Jolla, CA
BS in Structural Engineering with a focus in Aerospace Structures; GPA: 3.22 *Sep. 2026 – May 2028*

- **Relevant Coursework:** Mechanics of Materials, Dynamics, Thermodynamics, Statics, Computer-Aided Engineering, Circuits, Vibrations (Fall 2026), Statistics (Fall 2026), Structural Analysis I/II (Fall 2026), Fluid Mechanics (Winter 2026)

Arizona State University – Ira A. Fulton Schools of Engineering Tempe, AZ
Transferred to UCSD *Aug. 2024 – May 2026*

Technical Skills

CAD & Simulation: SolidWorks, AutoCAD, Autodesk Inventor (FEA), MATLAB (structural & dynamic modeling), 3D Printing
Manufacturing: CNC machining, metal lathe, precision cutting, drill press, power sander
Programming: Python, MATLAB, C++, Java | **Analysis**: Stress/fatigue analysis, failure mode analysis, root cause analysis

Work Experience

Commercial Metals Company (CMC Steel) June 2025 – Sep. 2025
Maintenance Reliability Engineering Intern *Mesa, AZ*

- Drafted industry standard technical drawings for critical components that will reduce possibilities of Liquid Metal Breakouts (LMB's) by 20 percent, cut out over 40 hours of downtime per month, and save over $100,000
- Diagnosed recurring structural failures on a high-cycle spool carrier using Autodesk Inventor FEA, performing fatigue and load path analysis to isolate root cause on a line processing 600,000+ tons of rebar and merchant bar annually
- Each failure caused 2–3 hours of downtime at $20,000+/hr; developed and submitted a corrective action package with load redistribution and material substitution recommendations to eliminate recurrence
- Learned simple fluid dynamics to increase pump efficiency by 30 percent, preventing compressors from overheating
- Authored multiple Root Cause Analyses (RCAs) on failed production equipment with the maintenance engineering manager, tracing failures to their originating cause and closing out corrective actions

Projects

Cryogenic Transfer Pipe Rack – Structural Design 2026

- Developed a single-bay steel pipe rack to support cryogenic propellant transfer lines (12" LOX, 8" CH4, 4" GN2) per ASCE 7 and AISC structural standards, sizing an 8-ft transverse span, 12-ft column height, and 20-ft bent spacing
- Generated a full structural drawing (S-101) in AutoCAD at 1/4"=1'-0" scale on a D-size layout, including title block, dimensioning, and member callouts to professional drafting standards

Autonomous Fire-Fighting Robot Sep. – Dec. 2024

- Engineered and built an autonomous fire-suppression robot with a 4-person team, programming temperature sensors, drive motors, and a water pump actuation system on an Arduino platform in C++, all integrated onto a custom-fabricated chassis to autonomously detect and extinguish a live open flame
- Completed 3 design iterations over a 12-week timeline to resolve sensor latency and actuation timing issues; final prototype repeatably detected and suppressed a live flame in a graded live demonstration

Chair Lift System – Mechanical Design Sep. 2024

- Built a motorized chair lift for a mobility-assistance application, performing torque and gear ratio calculations to size drivetrain components and validate structural margins under load; assembly CAD-modeled and structurally analyzed in Fusion 360
- Led a team of 4 through fabrication of a working prototype that successfully and repeatably lifted a 500g load; managed project schedule, task assignments, and risk tracking in Excel

Wooden Bridge Nov. 2022

- Designed and fabricated a balsa wood truss bridge to maximize load-to-weight ratio, applying truss geometry and joint design principles to distribute stress across the structure; withstood mechanical press loading in a graded competition setting

Extracurricular Activities

ASU Sun Devil Motorsports Tempe, AZ
Systems Team Member *Sep. 2025 – May 2026*

- Redesigned the brake fluid reservoir for a Formula SAE vehicle to accommodate an inverted master cylinder configuration, resolving packaging constraints while maintaining SAE compliance and structural integrity under braking loads
- Machined 20+ critical drivetrain and brake pedal components using CNC mill and metal lathe, contributing to an overall weight reduction of 1lb across manufactured parts; redesigned reservoir shipped on the competition car

ASU Solar Devils Tempe, AZ
Member *Feb. 2025 – May 2026*

- Developed MATLAB simulations to model and optimize mass distribution across a solar car chassis, identifying weight balance improvements to enhance vehicle handling and efficiency; created CAD models for braking system mounting brackets prioritizing mass minimization while maintaining structural rigidity under load

ASU American Society for Engineers Tempe, AZ
Member *Sep. 2025 – May 2026*

- Attended professional development workshops on resume writing and technical interviewing to strengthen industry readiness
- Participated in networking events connecting with mechanical/structural engineers in industry and alumni